Linux Processes Lab

Creating, managing, prioritising and terminating processes using the Linux command line

Viewing Processes with ps

1. Display your running processes and identify your shell process

```
ps
```

2. Display all the processes in long format

```
ps -ef
```

3. Display all the processes

```
ps -aux
```

Questions

a) What is the difference between the last two, which displays more processes?
b) What is the PID of the process that starts everything? What time did it start?
c) What is the PID of the process that is running ps? What time did it start?

Creating a Process with sleep

4. Start a process that runs for several minutes. and see its PID

```
sleep 300 &
ps -ef | grep sleep
```

Monitoring Processes with top

5. View CPU and memory usage in real time, and see what process is using the most resources

```
top
```

Background Processes

6. Start a process in the background and identify its PID.

```
sleep 600 &
jobs
```

7. Move jobs between foreground and background.

```
sleep 500
Ctrl+Z
bg
jobs
fg
```

Terminating Processes

8. Terminate a sleep process a process gracefully, you will need to create it and find its PID

```
sleep 600 &
jobs

kill PID
```

9. If you wanted to do it forcefully, use -9

```
sleep 600 &
jobs

kill -9 PID
```

Using nice

10. Start a low-priority process.

```
nice -n 10 sleep 300 &
```

11. Start a higher priority process.

```
nice -n 5 sleep 300 &
```

Later we will create a Bash script with an output and run it twice with two different prioities

Process Tree

12. Examine parent and child process relationships.

```
pstree
pstree -p
```

Process Signals

You can use kill to send different signals to a running process. What do you think cont and stop do?

```
kill -SIGSTOP PID
kill -SIGCONT PID
```